

AD Lab Notes

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These are lab notes from building an Active Directory environment using Windows Server 2022 and Windows 10 clients in VirtualBox. The document records both successful configurations and troubleshooting performed during the lab.

All passwords shown in this document are lab-only credentials used in an isolated virtual environment.

Setup

1. Installed virtual box
2. install Windows server 2022 standard evaluation desktop experience
 - user default: administrator
 - password: Longpassword123
 - 2 cores
 - 6000 mb ram
 - 50 GB storage
 - turned “Insert Guest Additions” off
3. install Windows 10 Clients 1
 - PC name is ‘Client 1’
 - there no password
 - username for client 1 is ‘User Client 1’
 - made the internet be internal only on the same one the server on
4. Clone Windows 10 Clients 1 for client 2
 - PC name is ‘Client 2’
 - also client 2 has no password
 - username for client 2 is ‘User Client 2’
5. changed the ram of the client 1 and 2 from the default 2000 to 4096 ram.
6. In Windows Server,
 - 1) I went to control panel
 - 2) network and internet
 - 3) network and sharing center
 - 4) change adapter settings (it on the left hand side)
 - 5) right click ethernet and click properties
 - 6) click internet protocol version 4, and properties
 - 7) changed the IP address to use a specific one instead of automatic, 192.168.56.10
 - 8) change subnet to 255.255.255.0
 - 9) change preferred DNS to 127.0.0.1
- 7: in the server manager
 1. click manager, then ‘add roles and features’
 2. make sure it ‘role-based or feature-based installation’
 3. there was only one server option so picked that one.

4. Checked the 'active directory domain services' and added the features
5. touch the notification flag
6. click "Promote this server to a domain controller"

8: domain controller settings

1. add a new forest
2. root domain name "adlab.local"
3. password "Longpassword123"
4. NetBIOS auto generated the domain name "ADLAB"
5. Database folder "C:\Windows\NTDS"
6. log file folder "C:\Windows\NTDS"
7. SYSVOL folder "C:\Windows\SYSVOL"

9. In Windows Client 1 and 2, I went to control panel

1. network and internet
2. network and sharing center
3. change adapter settings (it on the left hand side)
4. right click ethernet and click properties
5. click internet protocol version 4, and properties
6. changed the IP address to use a specific one instead of automatic,
Client 1 is 192.168.56.20, Client 2 192.168.56.30
7. change subnet to 255.255.255.0
8. change preferred DNS to 192.168.56.10

10. vm client1 and vm client2 domain joining

1. open cmd
2. use sysdm.cpl
3. select domain and put in 'adlab.local' into the domain area.
4. When prompted entered 'ADLAB\ADMINISTRATOR ' with password
Longpassword123
5. to double check logged in with other accounts, and enter 'ADLAB\ADMINISTRATOR '
with password Longpassword123

11: making Organizational Unit stuff

1. on the server, do the server manager
2. Click tools and "active directory Users and Computers
3. right click on 'adlab.local' and click 'new' then 'Organizational Unit'
4. Make OU of: Users2, IT, HR
5. in User2 right click and click 'new' then 'user'
 - o First Name: John
 - o Last name: Smith
 - o Username: jsmith
 - o password: Password123!
 - o Of note: to login you need to do adlab\jsmith
6. verified it worked by having both Server and Client1 running, and logging into jsmith
using Client1.

Disabling users.

open server and client 1

1. logged into john with client 1 and made a file

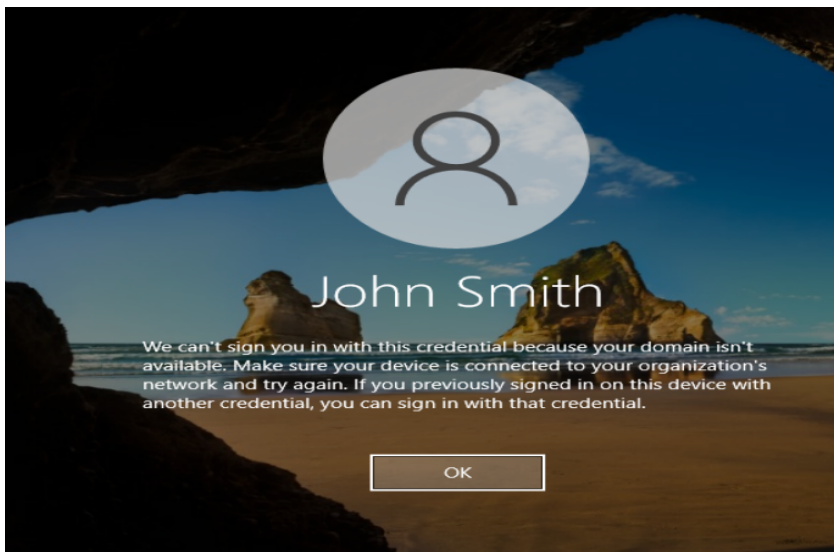
3. on server disabled john

- client 1 was not automatically signed out

4. after manually signing out client 1, attempted to login john account again and was told it was disabled.

- Tried pausing client 1, enable john on server, pause server, unpause client 1, and tried signing into john in client 1, took a while to load but got a error message (see picture)

5. after unpausing the server, client 1 can log back in as john, and the file is still there.



Adding Account lockout

in windows server

go to tools, and then group user policies

then Open Group Policy Management

Forest -> Domains -> adlab.local

Right clicked on 'Default Domain Policy' click edit

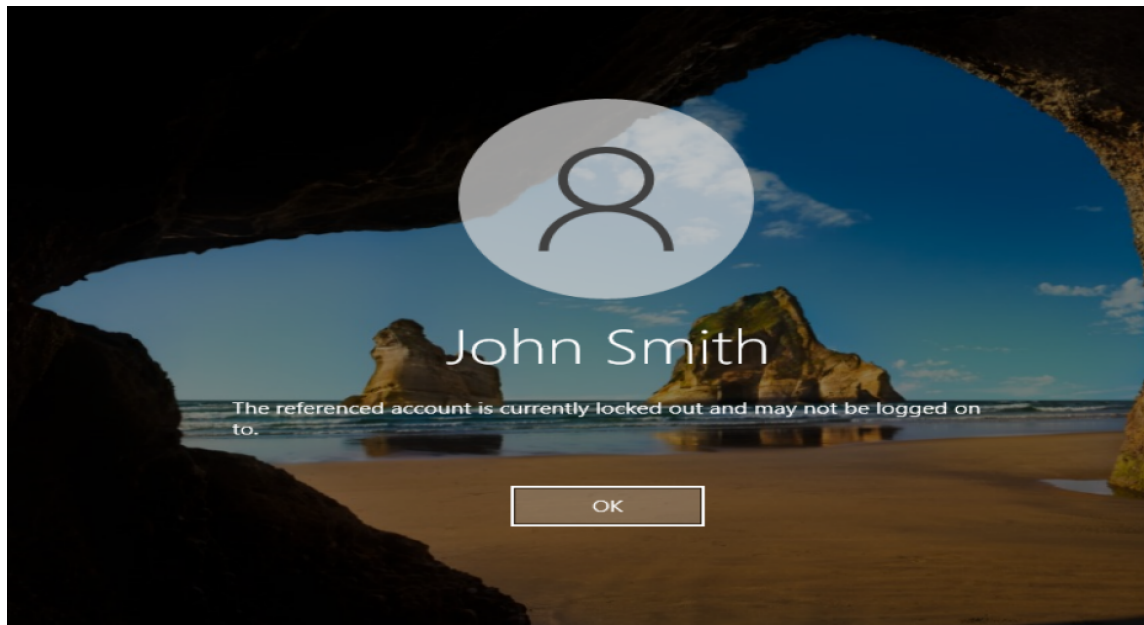
then another file chain.

Computer Configuration -> Policies -> Windows Settings -> Security Settings -> Account Policies

→ Account Lockout Policy

then 'account lockout threshold' and set it to 5 attempts, and 30 mins for lockout.

Incorrectly logged into client 1 john and got a lockout message



Unlocking accounts

server

tools → active directory users and computers

jsmith right click → properties, click the account tab, click the checkbox saying unlock account, click yes.

The screenshot shows the 'John Smith Properties' dialog box with the 'Account' tab selected. The 'User logon name' is 'jsmith' and the domain is '@adlab.local'. The 'User logon name (pre-Windows 2000)' is 'ADLAB\jsmith'. The 'Unlock account' checkbox is checked, with a message stating 'This account is currently locked out on this Active Directory Domain Controller.' The 'Account options' section includes checkboxes for 'User must change password at next logon', 'User cannot change password', 'Password never expires' (checked), and 'Store password using reversible encryption'. The 'Account expires' section has 'Never' selected. The 'OK' button is highlighted with a blue border.

Member Of	Dial-in	Environment	Sessions
Remote control	Remote Desktop Services Profile		COM+

General Address **Account** Profile Telephones Organization

User logon name:
jsmith @adlab.local

User logon name (pre-Windows 2000):
ADLAB\jsmith

Logon Hours... Log On To...

☒ Unlock account. This account is currently locked out on this Active Directory Domain Controller.

Account options:

- ☐ User must change password at next logon
- ☐ User cannot change password
- ☒ Password never expires
- ☐ Store password using reversible encryption

Account expires

☒ Never

☐ End of: Saturday, June 20, 2026

OK Cancel Apply Help

Reset password

Verified password “Password123!” Works on john

Click tools and “active directory Users and Computers

User2 → John Smith → right click → reset password

reseted password as “Newpass321!”

After password reset, john was not automatically logged off.

Tested old password, worked for a minute or two, followed by it no longer working
tested new password, and it worked.

Force password reset

set the “user must change account next login”

Click tools and “active directory Users and Computers

User2 → John Smith

right click → properties → account

unchecked account never expires

check the “user must change password next login” box

does not automatically sign client 1 out of john.

Logged out of john and tried logging back in.

Got a popup saying password must be changed before logging in

changed password to: New2pass321!

Add to groups

Add John to HR_Group

1. Open Active Directory Users and Computers
2. Find John Smith
3. Right-click John Smith → Click Properties
4. Open the Member Of tab
5. Click Add
6. Type: HR_Group
7. Click Check Names
8. Click OK
9. Click Apply
10. Click OK

Verify Membership

1. Open John Smith Properties
2. Go to Member Of
3. Confirm:
 - Domain Users
 - HR_Group

Test Access

1. Sign into Client1 as John.
2. Open: \\192.168.56.10\HR
3. Verify:
 - HR folder opens.
 - Files can be created/modified.

Remove John from HR_Group

1. Open John Smith Properties
2. Go to Member Of
3. Select HR_Group
4. Click Remove
5. Click Apply
6. Click OK

Verify Removal

1. Reopen John Smith Properties
2. Go to Member Of

3. Confirmed only: Domain Users

Test Loss of Access

1. Signed out as John.
2. Sign back in.
3. Attempt to access: \\192.168.56.10\HR
4. Verify access is denied or no longer allows modification.

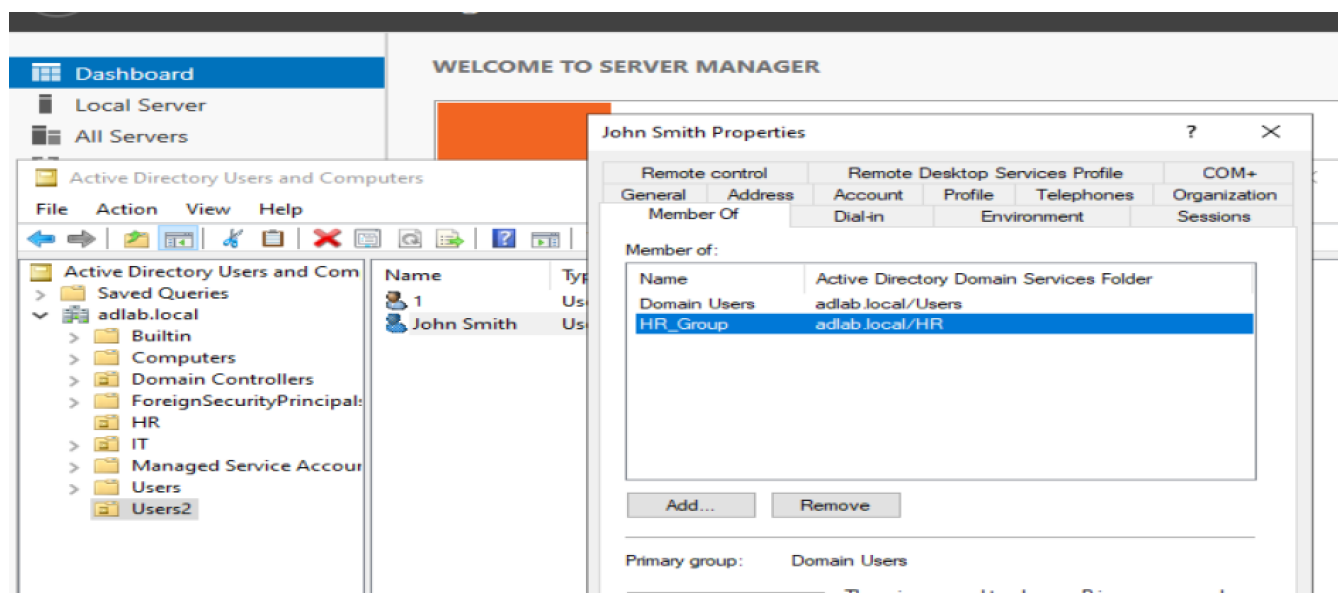
Troubleshooting:

1: well the HR_Shared folder advanced shared had the HR, and the security tab had the HR_Group with permissions, John could not save the document. After some searching in the advanced shared tab, I didn't add permissions for everyone to make edits.

2: I didn't see that ADLAB\Users were granting john permission, but I couldn't remove John from ADLAB\Users or edit the permissions because of inheritance, so I had to remove inheritance to be able to remove the permissions

side note:

If someone is logged in, and then added to a group the change will update on their account, however, if someone is removed from the group, it will not update until they sign out and sign back in.



Wallpaper Policy

server

click Tools → Group Policy Management

Forest → Domains → adlab.local

Right-click Users2 and choose “Create a GPO in this domain, and Link it here” name it “Wallpaper Policy” and click ok

right-click Wallpaper Policy and select Edit

added a wallpaper folder and changed it settings to share, added a jpg to the folder.

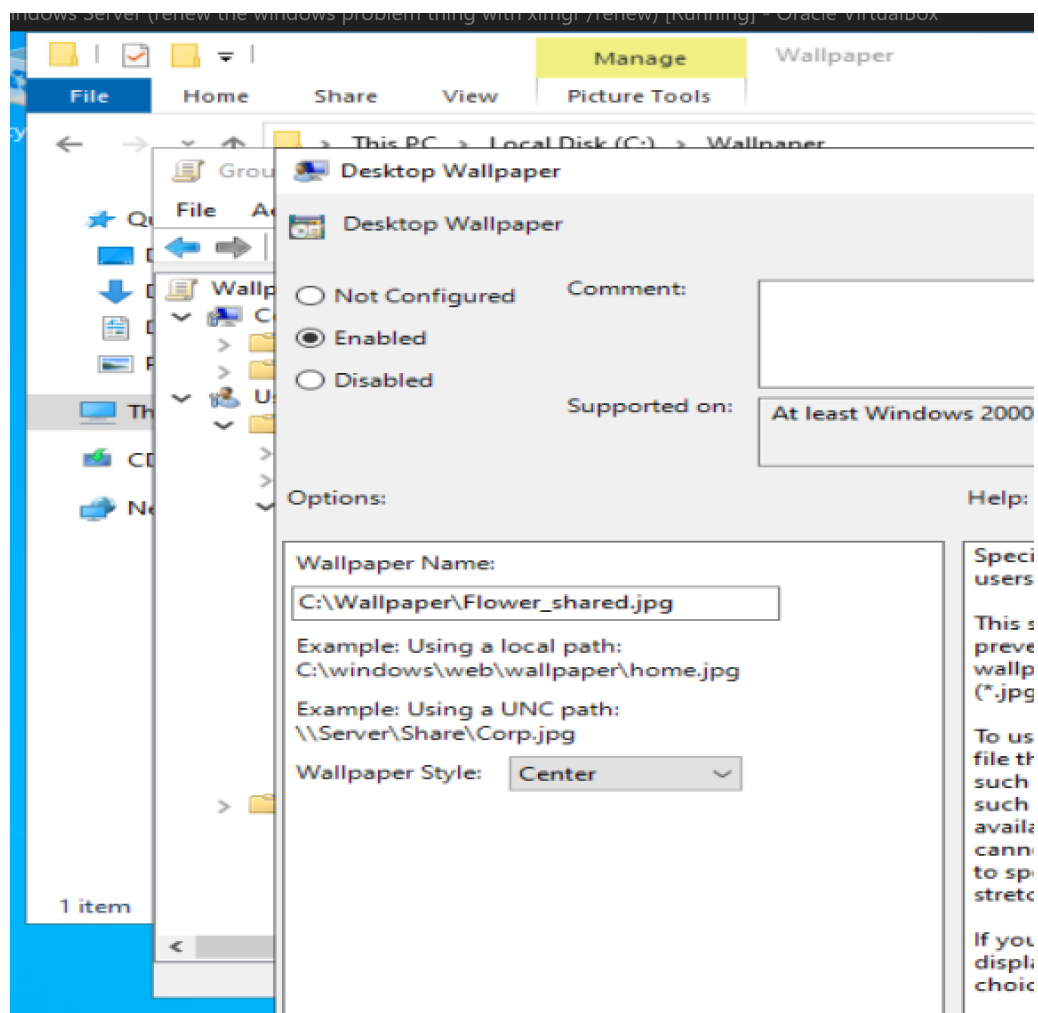
Did gpupdate /force to force a update.

didn't change until logged out and then back in.

tried doing C:\Wallpaper\shared_flower.jpg but that doesn't work because John tries looking at their own C drive, needed to make the path to the wallpaper be

[\\HostName\Wallpaper\shared_flower.jpg](#)

where hostname is the result of doing “hostname” in the cmd of the server.



Disable control panel for users.

Server

Server Manager → Tools → Group Policy Management

Forest → Domains → adlab.local

Right-click Users2 → Create a GPO in this domain, and Link it here

Named it “Disable Control Panel”

Right click “Disable Control Panel” and click Edit

User Configuration → Policies → Administrative Templates → Control Panel

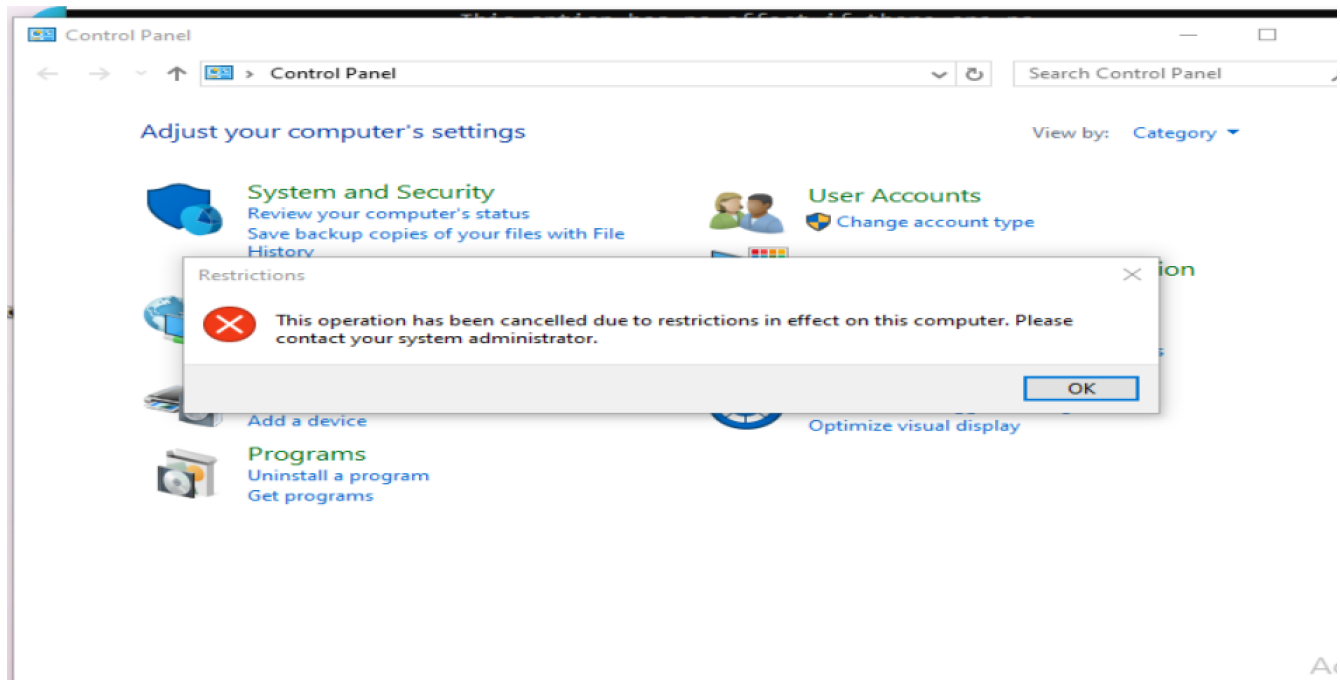
Prohibit access to Control Panel and PC settings

Set it to Enabled

Click Apply and OK.

After running I ran gpupdate /force

control panel became immediately unable to be used. Even the already open control panel wouldn't let me interact with any of the controls on it.



Prohibit access to Control Panel and PC settings

Previous Setting Next Setting

☐ Not Configured Comment:

☒ Enabled

☐ Disabled

Supported on: At least Windows 2000

Options:

Help:

Disables all Control Panel programs and the PC settings app.

This setting prevents Control.exe and SystemSettings.exe, the program files for Control Panel and PC settings, from starting. As a result, users cannot start Control Panel or PC settings, or run any of their items.

This setting removes Control Panel from:

- The Start screen
- File Explorer

This setting removes PC settings from:

- The Start screen
- Settings charm
- Account picture
- Search results

If users try to select a Control Panel item from the Properties item on a context menu, a message appears explaining that a setting prevents the action.

OK Cancel Apply

Disabling USB & Move PCs between OUs

Server

Open: Server Manager → Tools → Group Policy Management

Forest → Domains → adlab.local

Right-click Users2 → Create a GPO in this domain, and Link it here

Named it “Disable USB Storage”

User Configuration → Policies → Administrative Templates → System → Removable Storage Access

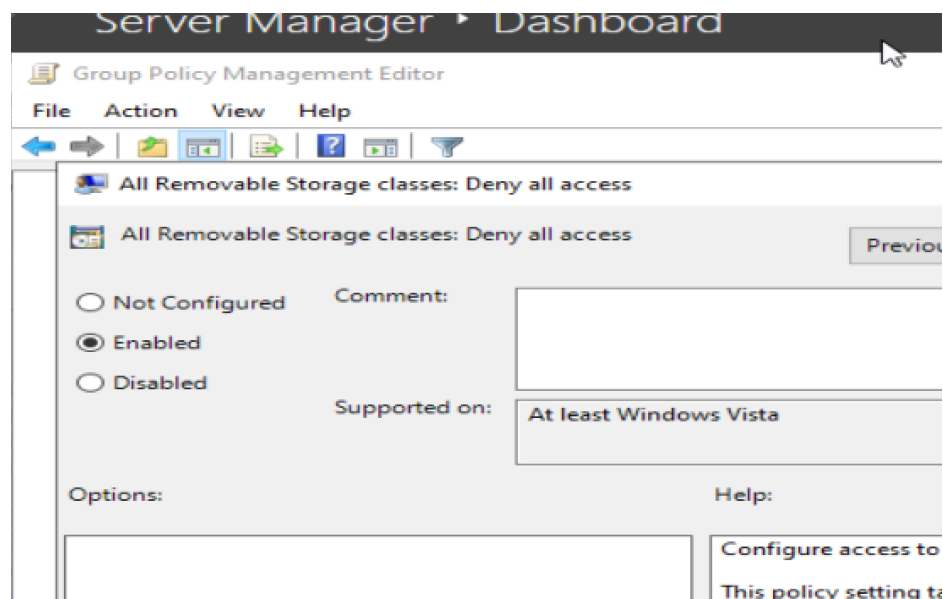
clicked “All Removable Storage classes: Deny all access”

Set it to Enabled.

Then on Client1:

- Run gpupdate /force
- Plug in a USB drive if you have one available to pass through VirtualBox

ran into a problem where despite verifying that GPO was enabled, and John was confirmed updated, the GPO still didn't run. After some troubleshooting discovered that the OU the GPO was attached to was only attached to a OU that only contained users, however, disable USB is a computer GPO, so I had to put the computers into their own OU and then link the disable GPO rule to it and it worked fine.



Mapping Drivers

Server

Server Manager → Tools → Group Policy Management

Forest → Domains → adlab.local

Right click Users2 → Create a GPO in this domain, and Link it here

Named it “Map HR Drive”

User Configuration → Preferences → Windows Settings → Drive Maps

Right-click inside the empty right pane.

New → Mapped Drive

Action Create

Location <\\192.168.56.10\hr>

Drive Letter: H

On Client1

Logged in as John.

Ran gpupdate /force

restarted computer

Then Tested

Open: the file explorer when logged in as John and saw the shared file.

Trouble I ran into:

Was not working when I had it as

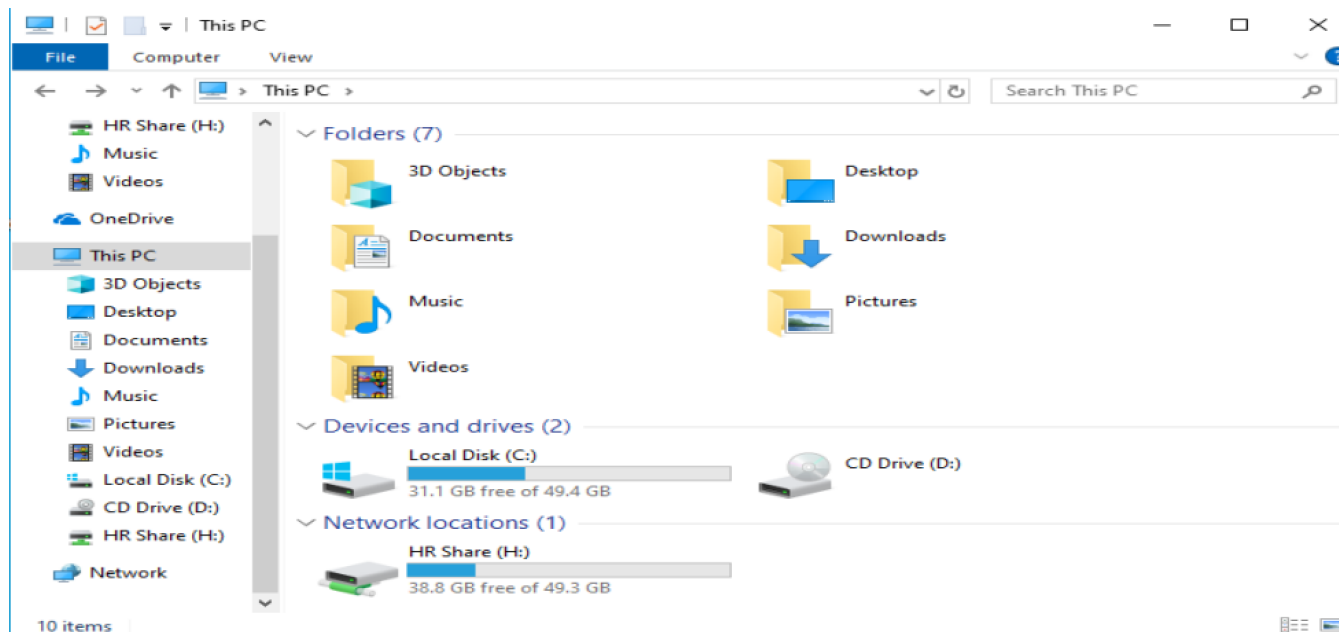
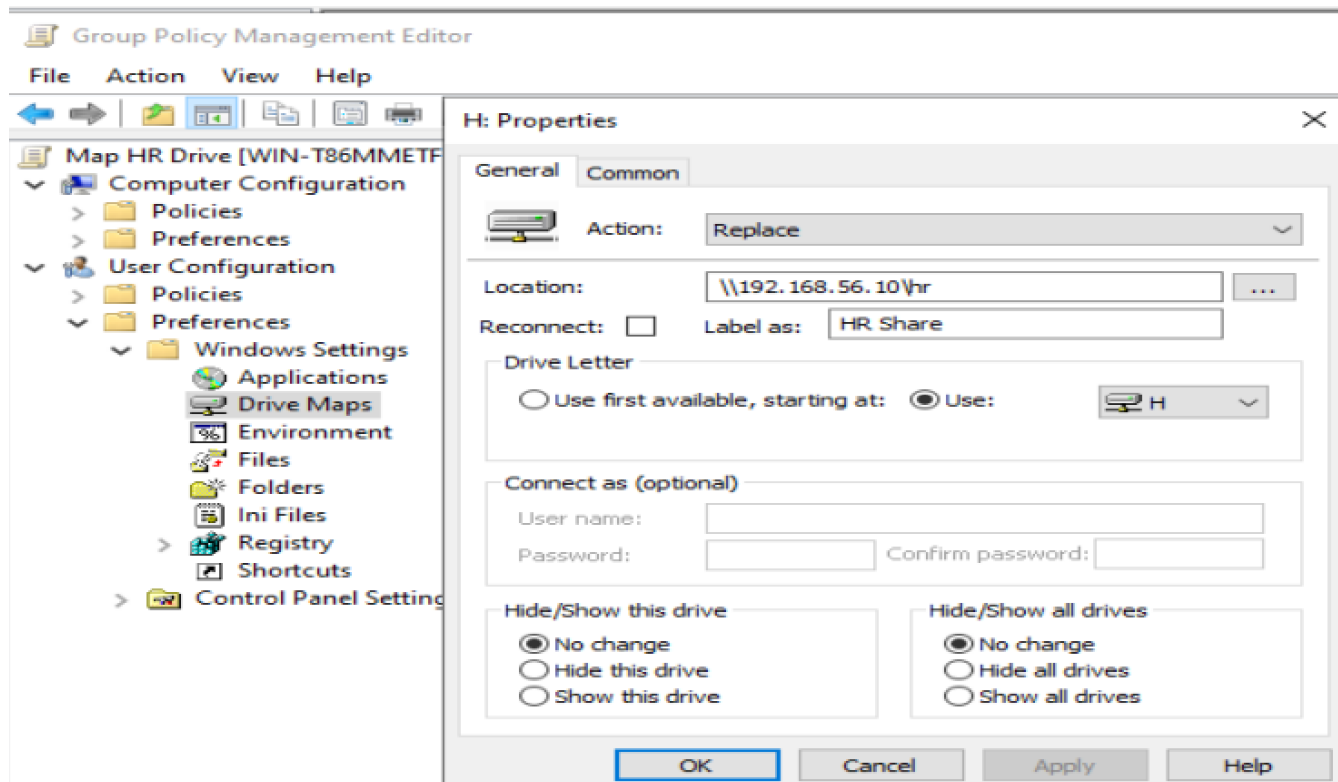
\\192.168.56.10\Share\HR_Shared

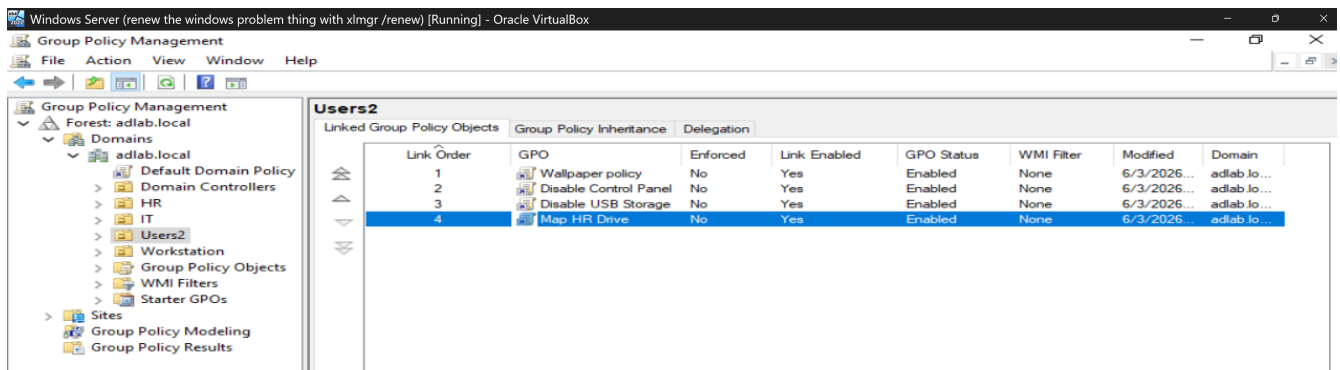
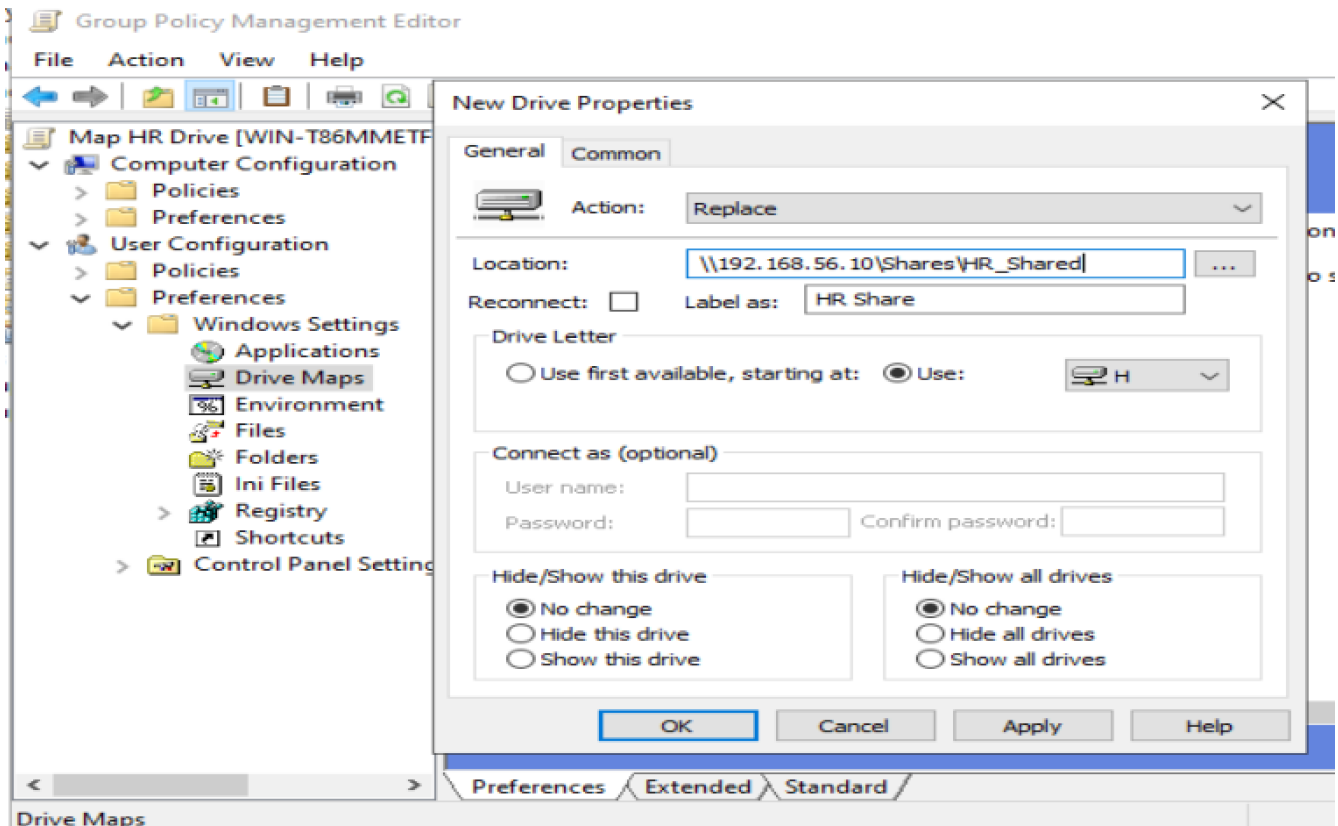
which is the names for the folders on the server

but works with

<\\192.168.56.10\hr>

something about this being the shared name of the folder, and that the name well shared might not be it name on the server.





Remove from domain

On Client1

Logged with a local account (User Client 1).

Run: sysdm.cpl

Open the Computer Name tab -> Change.

Verify current name as adlab.local

Selected Workgroup

named it "WORKGROUP"

Click OK.

Receive the message:

Welcome to the WORKGROUP workgroup.

Click through the remaining prompts.

Reboot Client1.

Verification

After reboot:

1. Logged in with User Client 1
2. Run: whoami
3. Verify the result shows: client1\User Client 1

Remote connection

Client1

1. Run: sysdm.cpl
2. Open the Remote tab.
3. Under Remote Desktop selected Allow remote connections to this computer
4. Click: Apply
5. Click: OK

Server

1. Run: mstsc
2. In the Computer field enter CLIENT1
3. Click: Connect

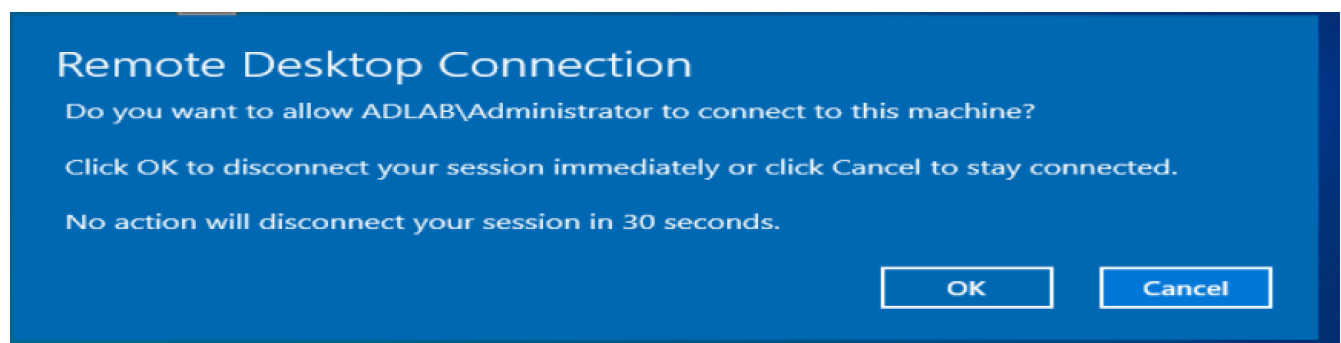
Because Client1 already had John logged in locally, Windows displayed a warning that another user was currently signed in and would be disconnected if the remote connection continued.

1. Click: Yes
2. Continue the connection.

Verification

After connecting:

1. Client1 desktop appeared inside the Remote Desktop window on the Server.
2. Opened File Explorer through the remote session.
3. Opened C:\Users\John
4. Verified access to files stored in John's profile.



Broken domain

Client1:

run ipconfig /all

Verified DNS as 192.168.56.10

Client1:

Open: Control Panel → Network and Internet → Network and Sharing Center → Change adapter settings

Right-click Ethernet->Properties->Internet Protocol Version 4 (TCP/IPv4)->Change Preferred DNS Server

from:

192.168.56.10

to:

8.8.8.8

Clicked OK

Test 1 – Domain Name Resolution

On Client1:

ping adlab.local

Result: Could not find host adlab.local

Test 2 – Connectivity by IP Address

Client1:

ping 192.168.56.10

Result: Reply from 192.168.56.10

Test 3 – Network Share by Name

In File Explorer \adlab.local

Result: Windows cannot access \adlab.local

Test 4 – Network Share by IP

In File Explorer \192.168.56.10

Result: Share was reachable.

After entering credentials, the HR shared folder could be accessed.

Fix

Return to:

Control Panel → Network and Internet → Network and Sharing Center → Change adapter settings → Ethernet Properties → IPv4 Properties

Change DNS from: 8.8.8.8

back to: 192.168.56.10

Click OK.

Clear DNS Cache

run ipconfig /flushdns

Verification

Test 1

ping adlab.local

Result: Successful replies received.

Test 2

In File Explorer \adlab.local

Result Server shares accessible again.

Disabled NIC

Client1

run ping 192.168.56.10

result: Reply from 192.168.56.10

Disable the NIC

Client1

Run: ncpa.cpl

Right-click Ethernet → Disable

Test 1 – Connectivity Test

run ping 192.168.56.10

Result: PING: transmit failed. General failure.

Test 2 – Network Configuration Check

Run: ipconfig

Result: it only said “Windows IP Configuration” and displayed nothing else.

Fix

Client1:

Run ncpa.cpl

Ethernet → Right-click → Ethernet → Enable

Wait a few seconds for Windows to initialize the adapter. (attempted to test too early and it failed)

Verification

Test 1 – Check Network Configuration

Run: ipconfig

IPv4 Address: 192.168.56.20

Subnet Mask: 255.255.255.0

Test 2 – Connectivity Test

Run: ping 192.168.56.10

Result Reply from 192.168.56.10

Note:

The first ping immediately after enabling the adapter may fail while Windows reinitializes the interface. Subsequent pings should succeed once the adapter is fully operational.

Conclusion

The workstation's network adapter was intentionally disabled to simulate loss of network connectivity. The system was unable to transmit traffic and could not communicate with the Domain Controller. Re-enabling the NIC restored network functionality and communication with the server.